

CS223 Computer Architecture & Organization

DRAM Organization



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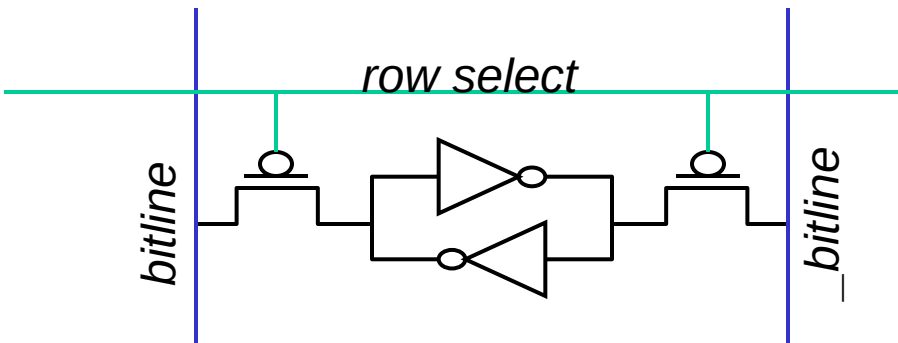
Associate Professor

Department of Computer Science & Engineering

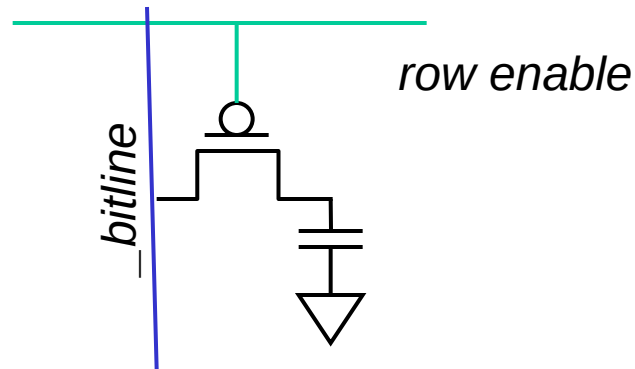
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SRAM vs DRAM

SRAM

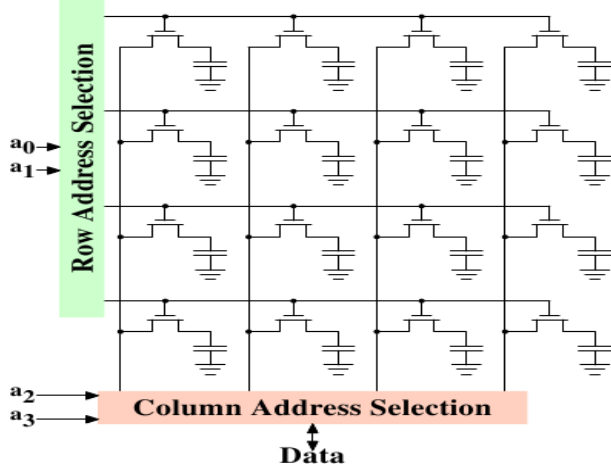


DRAM



DRAM (Dynamic Random Access Memory)

- ❖ Bits are stored as charges on capacitor
- ❖ B cell loses charge when read
- ❖ B cell loses charge over time
- ❖ A “flip-flopping” sense amplifier amplifies and regenerates the bitline, data bit is mux'ed out



DRAM vs SRAM

❖ DRAM

- ❖ Slower access (capacitor)
- ❖ Higher density (1T, 1C cell), Lower cost
- ❖ Requires refresh (power, performance, circuitry)
- ❖ Manufacturing requires putting capacitor and logic together

❖ SRAM

- ❖ Faster access (no capacitor)
- ❖ Lower density (6T cell), Higher cost
- ❖ No need for refresh
- ❖ Manufacturing compatible with logic process (no capacitor)

Principle of Interleaving

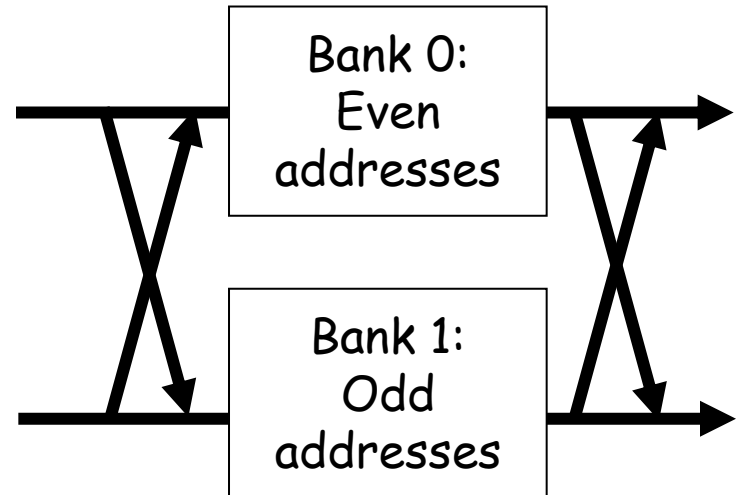
❖ Banking (Interleaving)

- ❖ **Problem**: a single monolithic memory array takes long to access and does not enable multiple accesses in parallel
- ❖ **Goal**: Reduce the latency of memory array access and enable multiple accesses in parallel
- ❖ **Idea**: Divide the array into multiple banks that can be accessed independently (in the same cycle or in consecutive cycles)
 - ❖ Each bank is smaller than the entire memory storage
 - ❖ Accesses to different banks can be overlapped
- ❖ **A Key Issue**: How do you map data to different banks?

Handling Multiple Accesses per Cycle

❖ Banking (Interleaving)

- ❖ Address space partitioned into separate banks
- ❖ No increase in data store area
- ❖ Bits in address determines which bank an address maps to
- ❖ Cannot satisfy multiple accesses to the same bank
- ❖ Crossbar interconnect in input/output
- ❖ **Bank conflicts** - Two accesses are to the same bank difficult to handle



Page Mode DRAM

- ❖ A DRAM bank is a 2D array of cells: rows x columns
- ❖ Sense amplifiers are kept in row buffer
- ❖ Each address is a <row,column> pair
- ❖ **Access to a closed row**
 - ❖ **Activate** command opens row (placed into row buffer)
 - ❖ **Read/write** command reads/writes column in the row buffer
 - ❖ **Precharge** command closes the row and prepares the bank for next access
- ❖ **Access to an open row**
 - ❖ No need for activate command

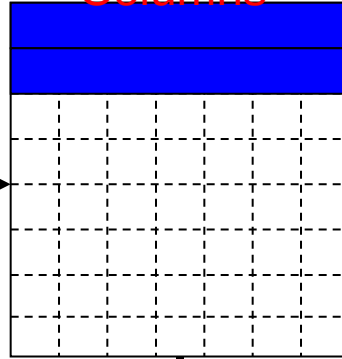
DRAM Bank Operation

Access Address:
(Row 0, Column 0)
(Row 0, Column 1)
(Row 0, Column 85)
(Row 1, Column 0)

Row address 0

Row decoder

Columns



Rows



Row Buffer **CONFLICT !**

Column address 5

Column mux

Data

DRAM Subsystem Organization

- ❖ Channel

- ❖ DIMM

- ❖ Rank

- ❖ Chip

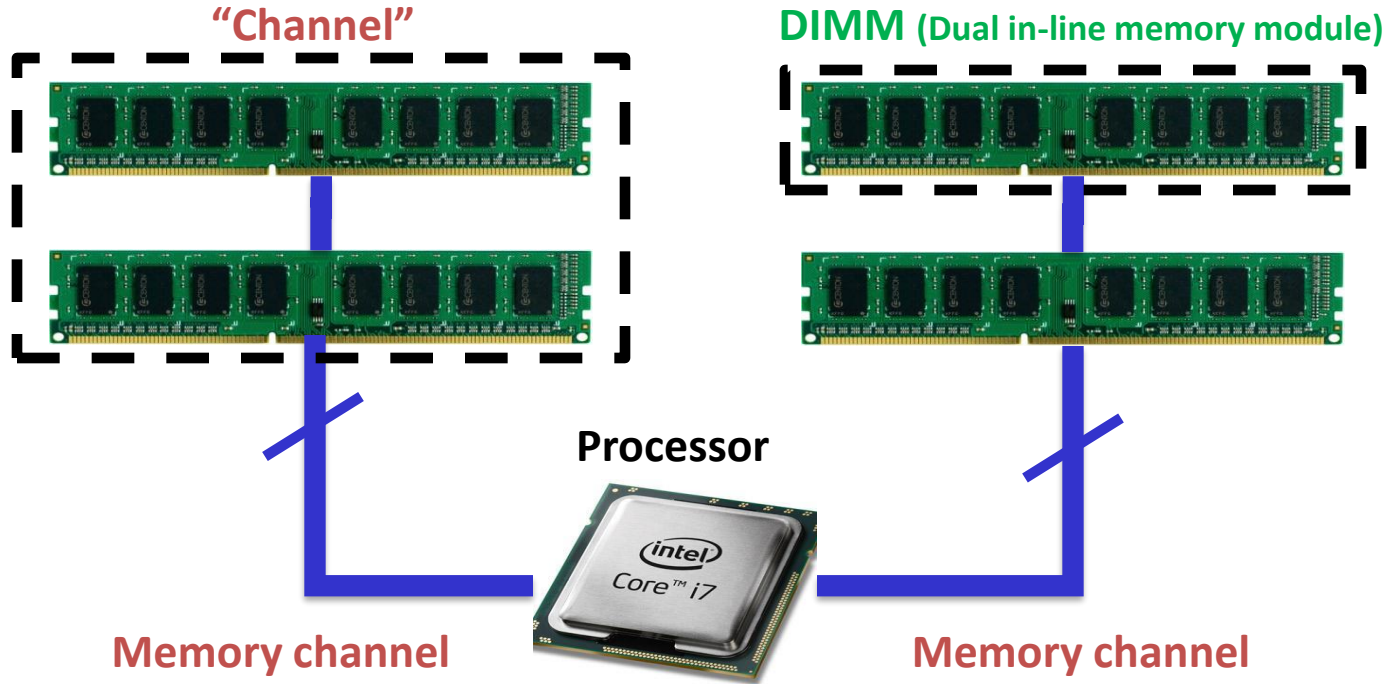
- ❖ Bank

- ❖ Row

- ❖ Column

- ❖ B-Cell

The DRAM subsystem



Breaking down a DIMM

DIMM (Dual in-line memory module)



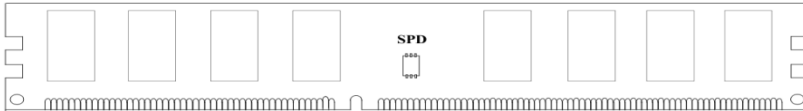
Side view

SIDE

4.00

Front of DIMM

Back of DIMM



Breaking down a DIMM

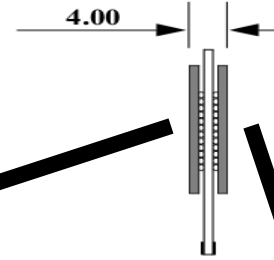
DIMM (Dual in-line memory module)



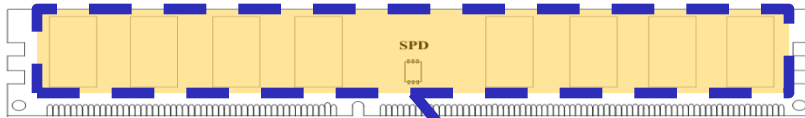
Side view

SIDE

4.00

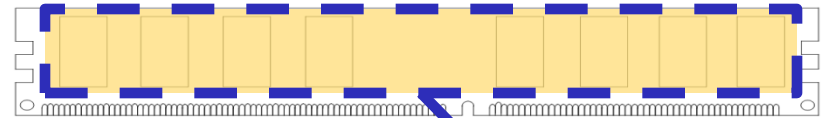


Front of DIMM



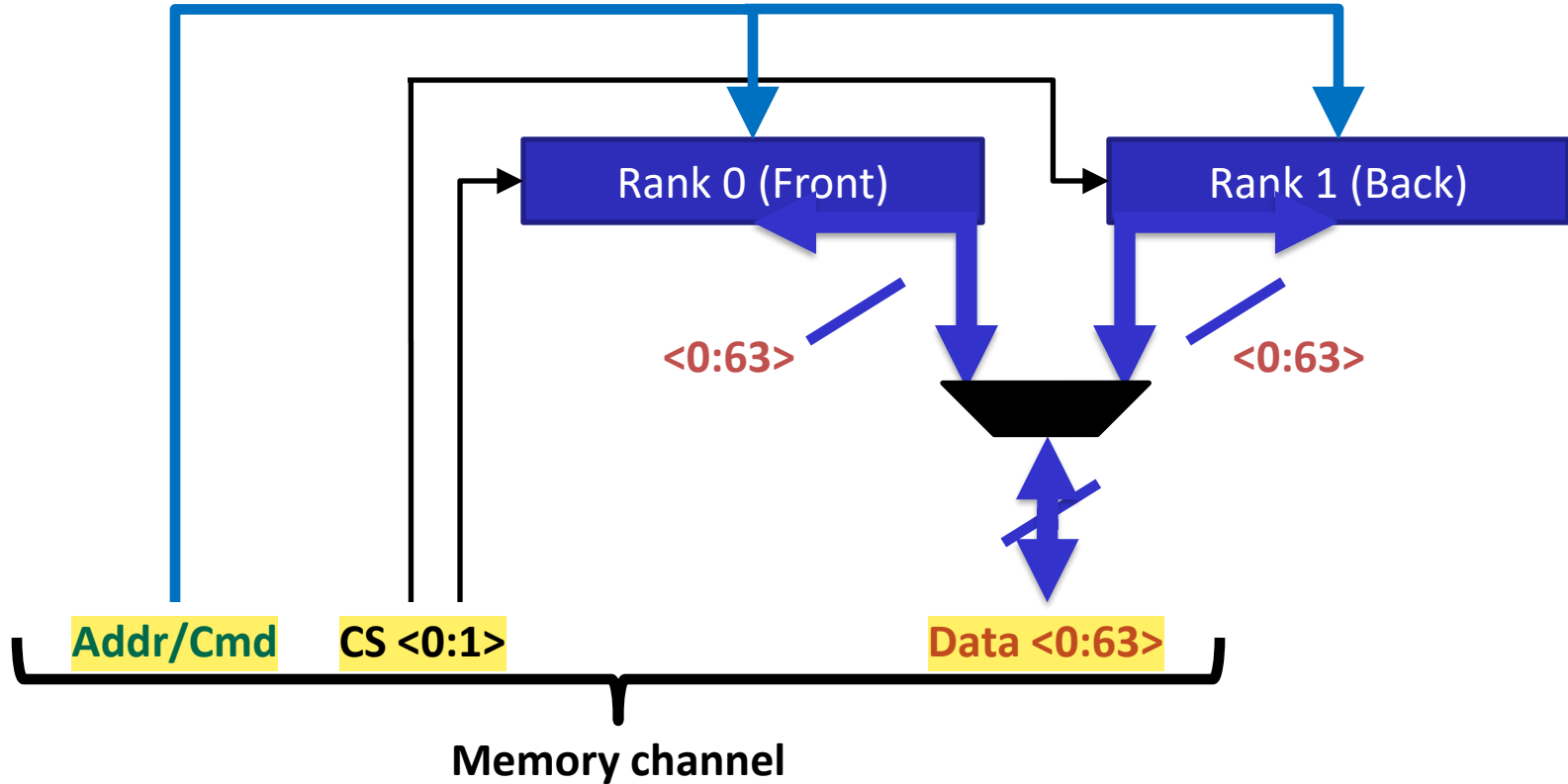
Rank 0: collection of 8 chips

Back of DIMM

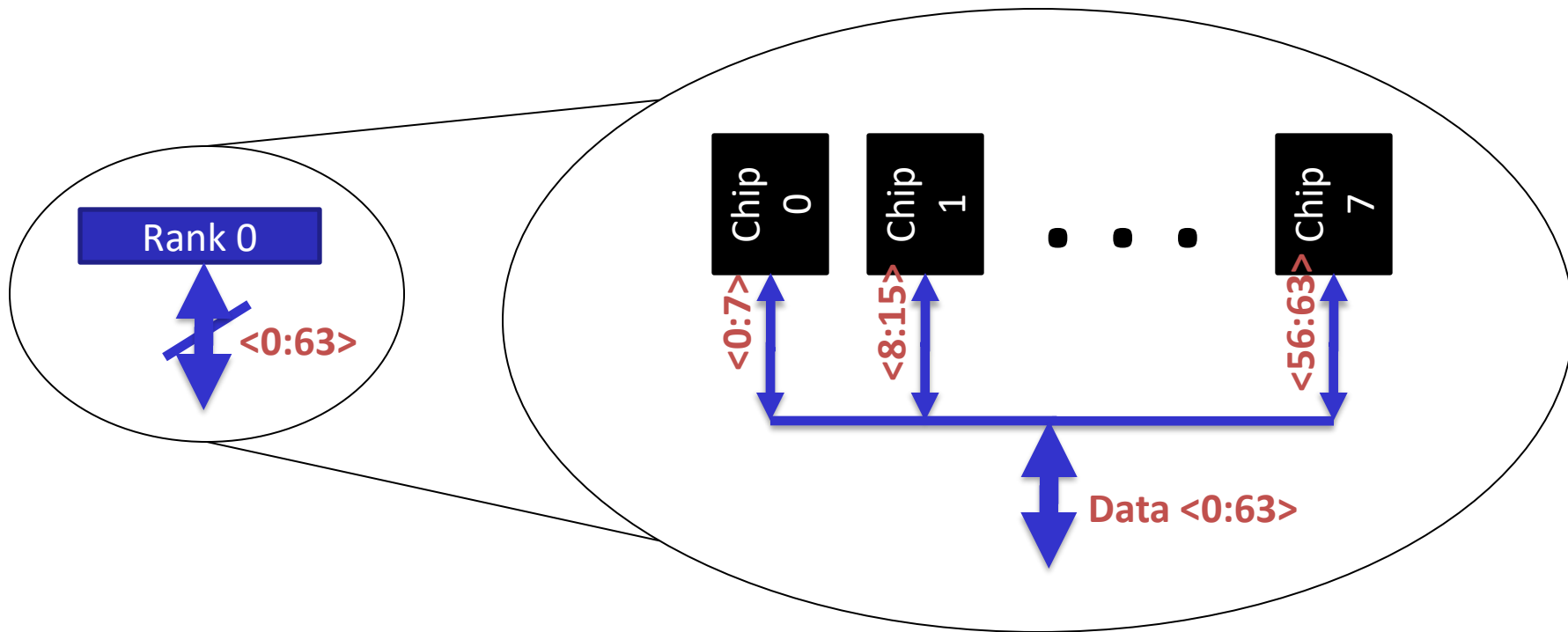


Rank 1

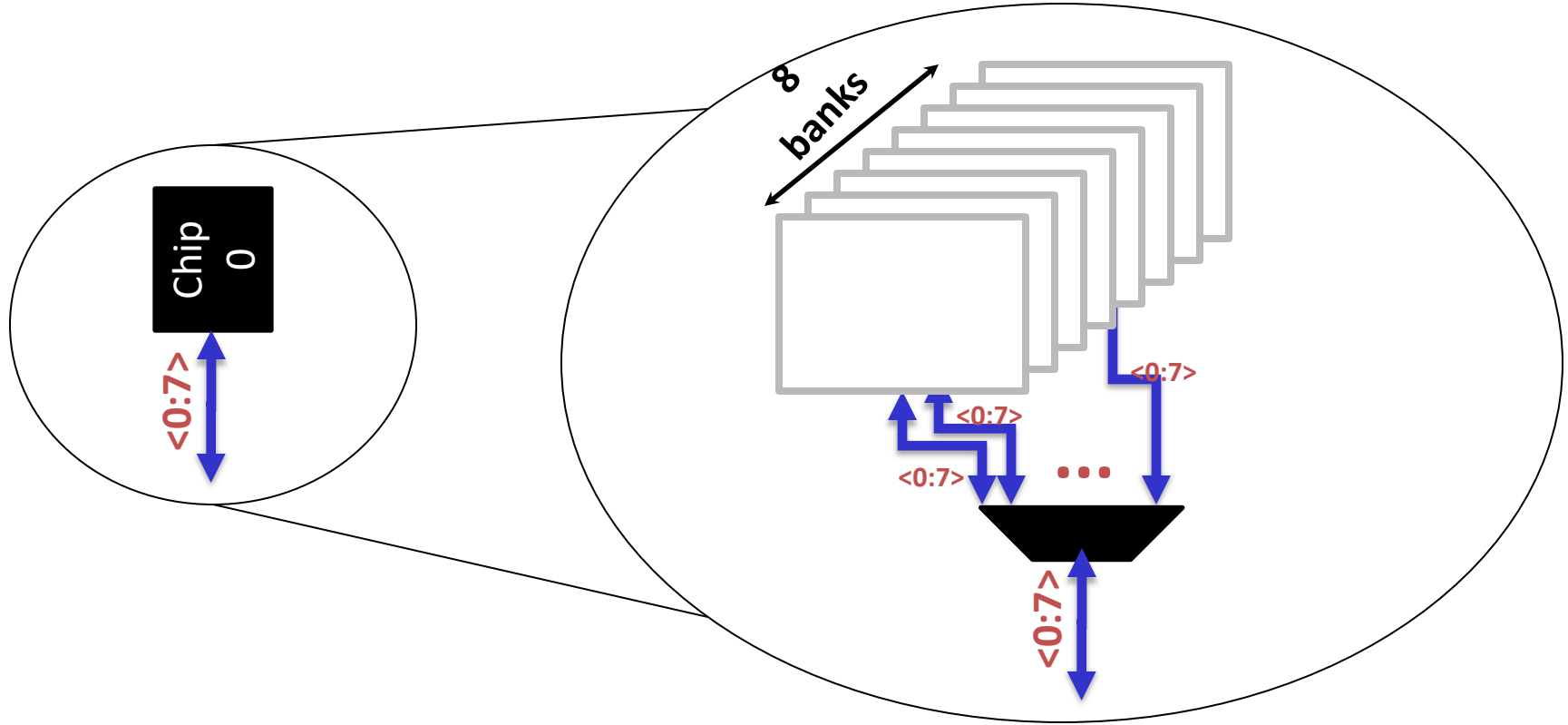
Rank



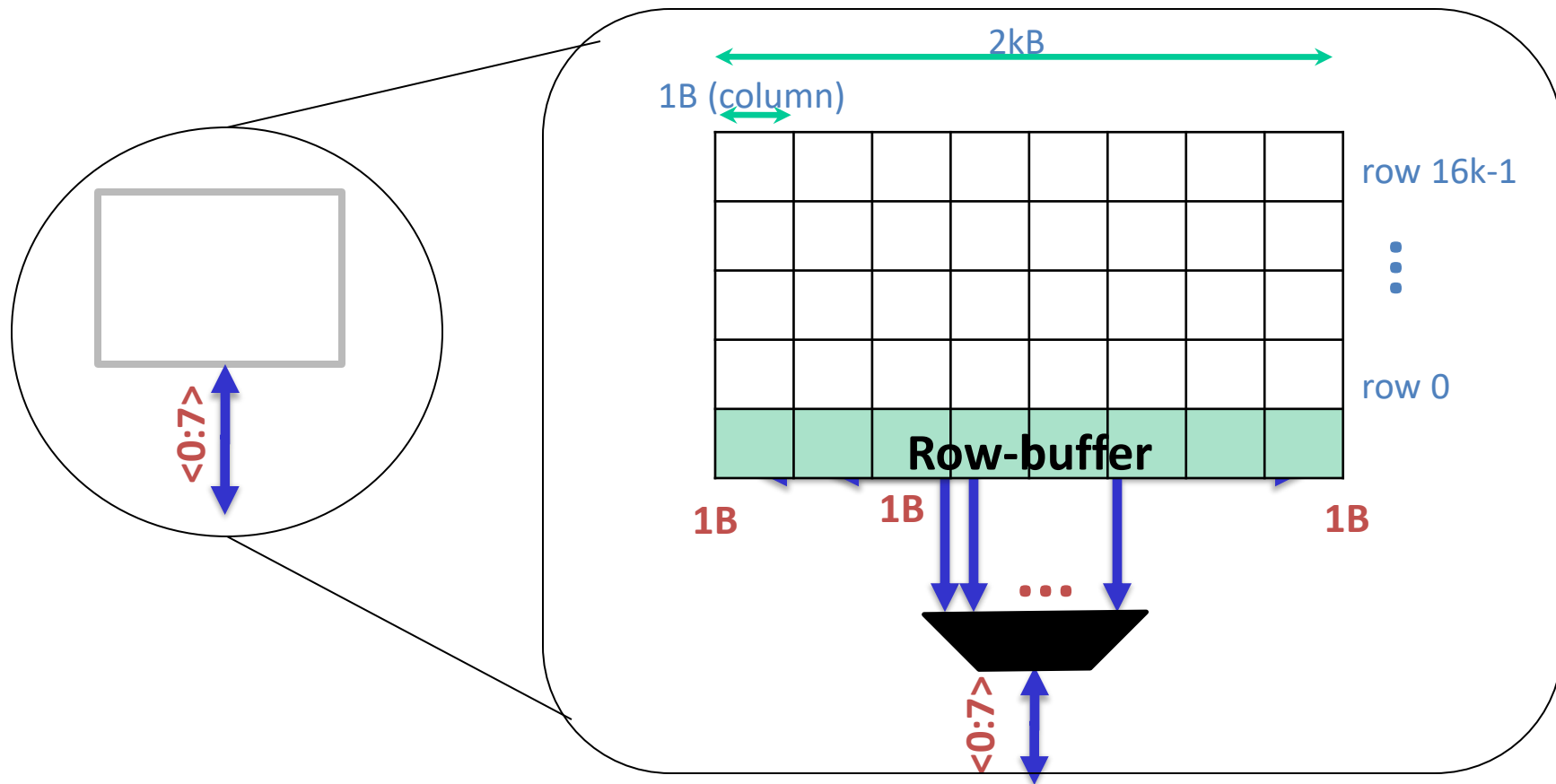
Breaking down a Rank



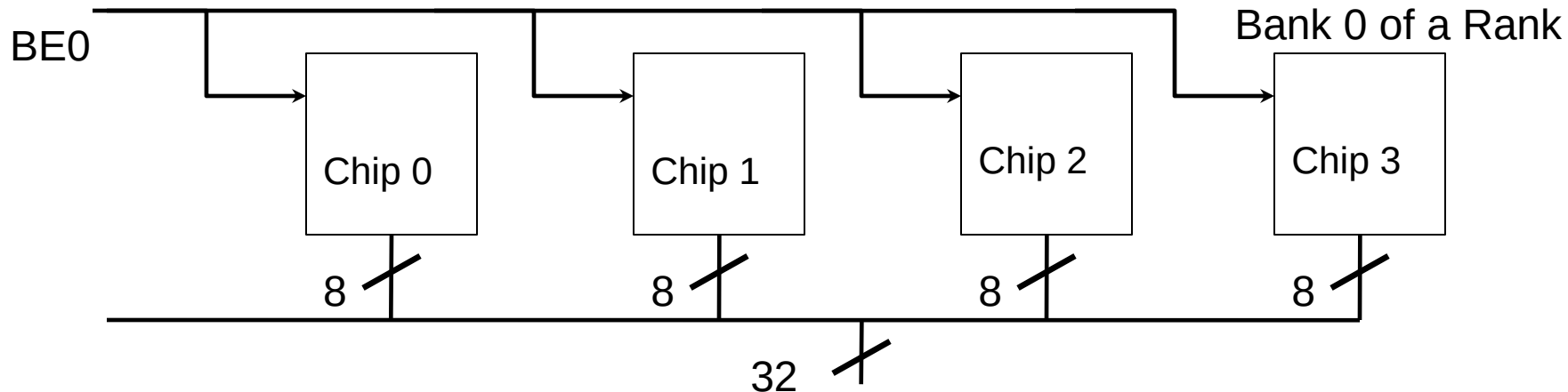
Breaking down a Chip



Breaking down a Bank



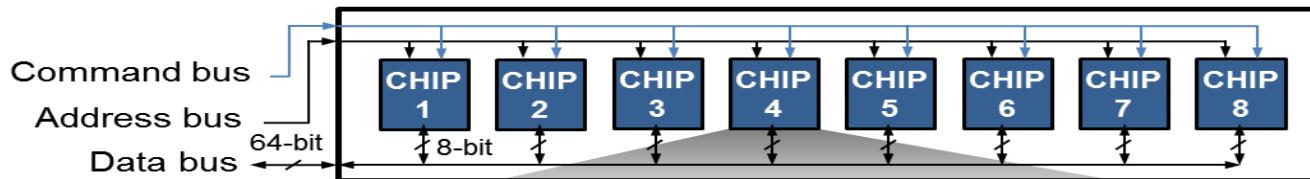
DRAM Rank



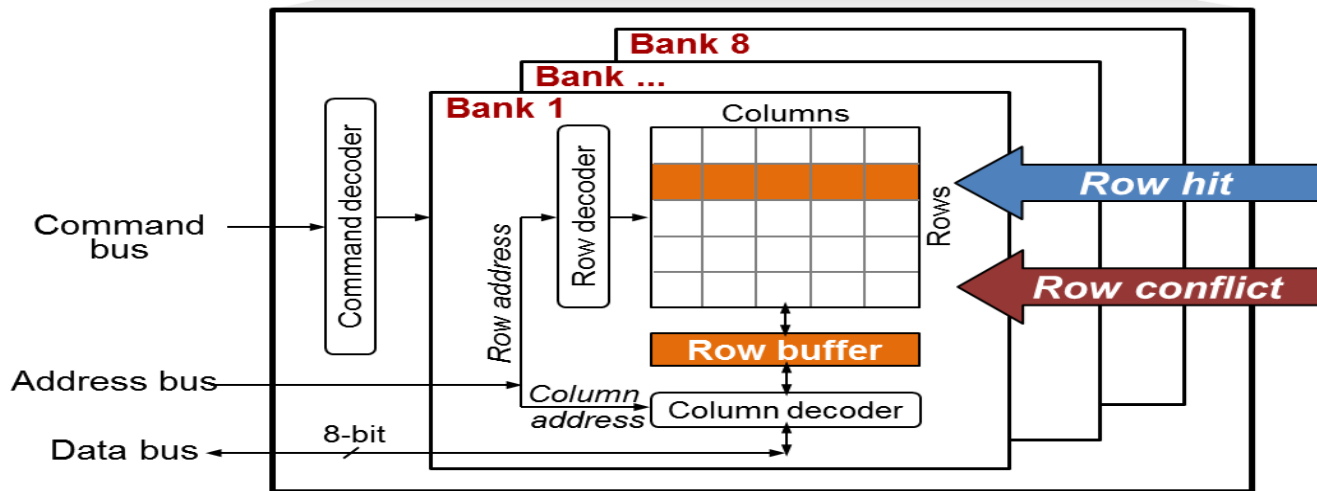
- ❖ Rank : A set of chips that respond to same command and same address at the same time but with different pieces of the requested data.
- ❖ Easy to produce 8 bit chip than 32 bit chip.
- ❖ Produce an 8 bit chip but control and operate them as a rank to get a 32 bit data in a single read.

DRAM Rank

DRAM Rank

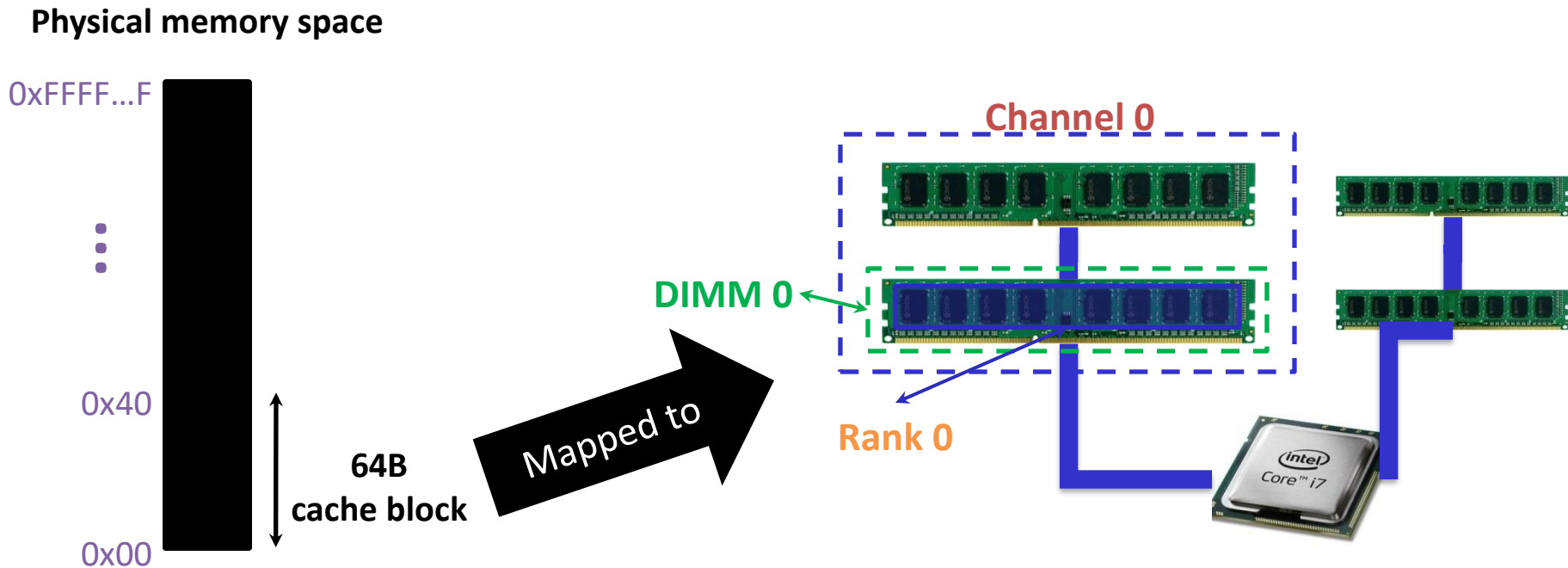


DRAM Chip

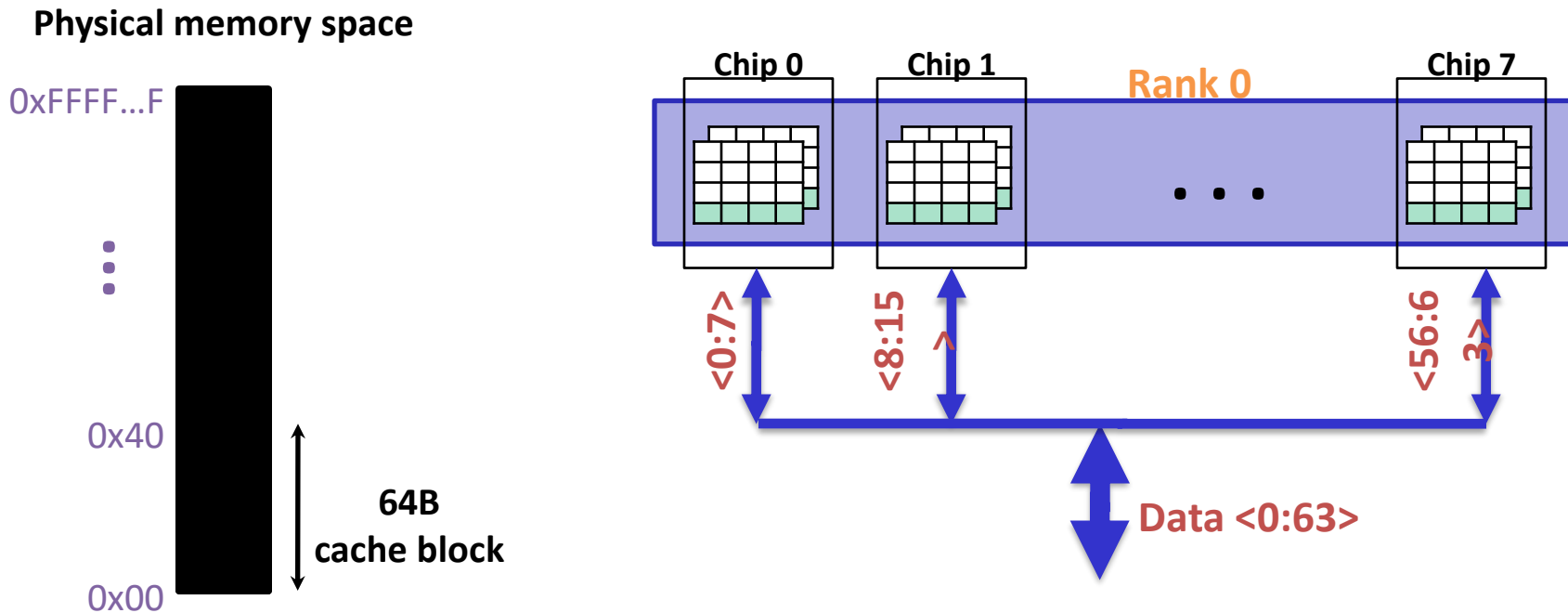


DRAM access latency varies depending on which row is stored in the row buffer

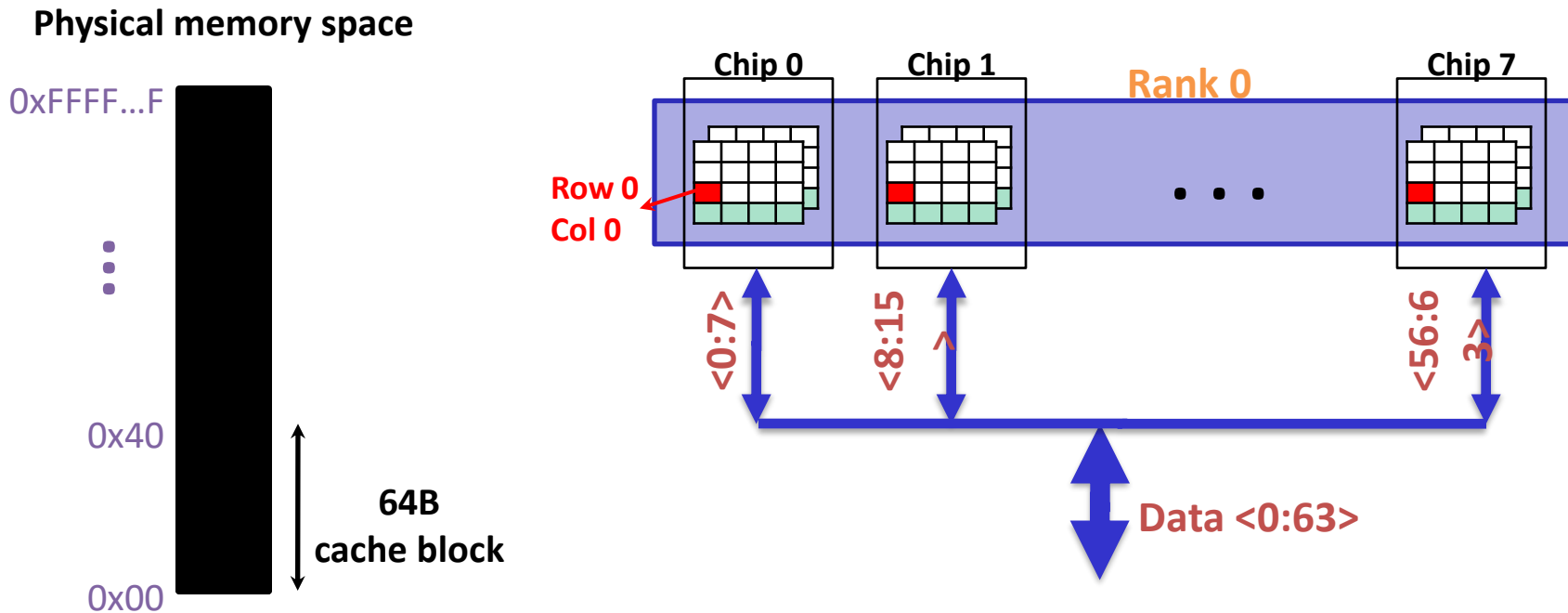
Transferring a cache block



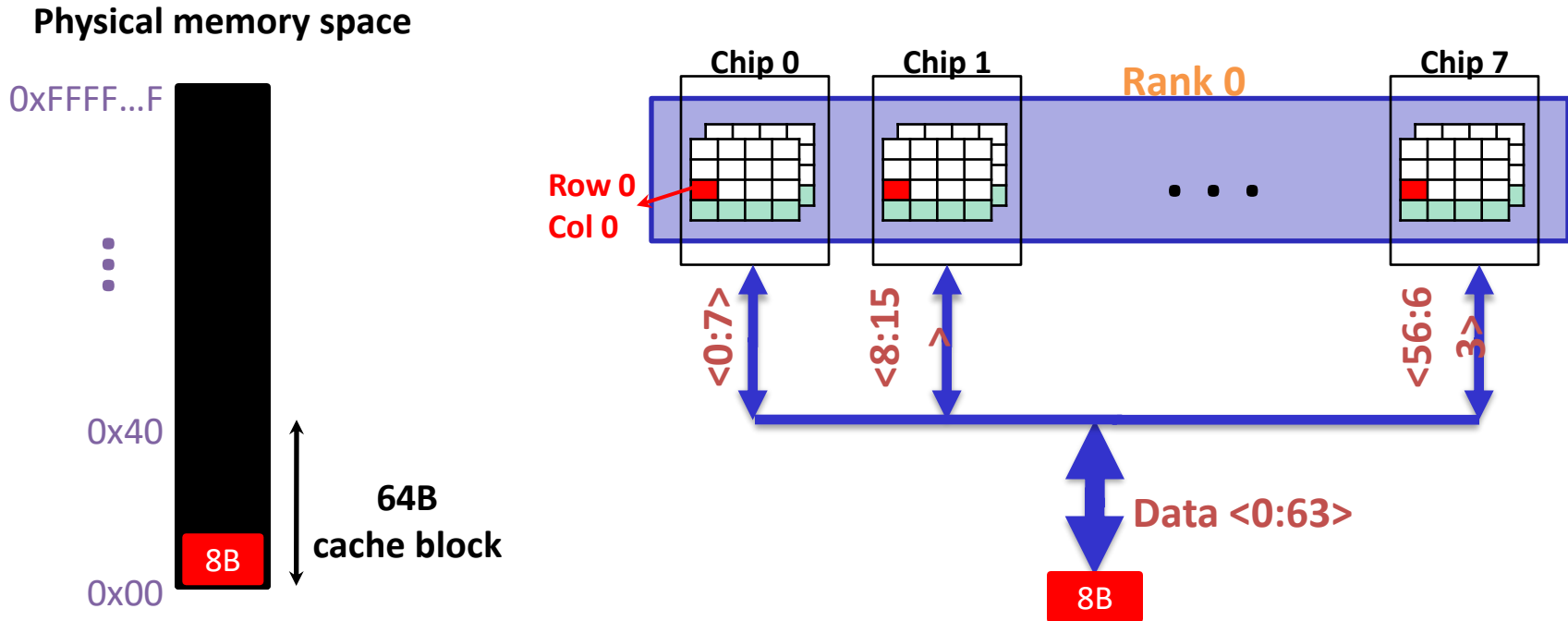
Transferring a cache block



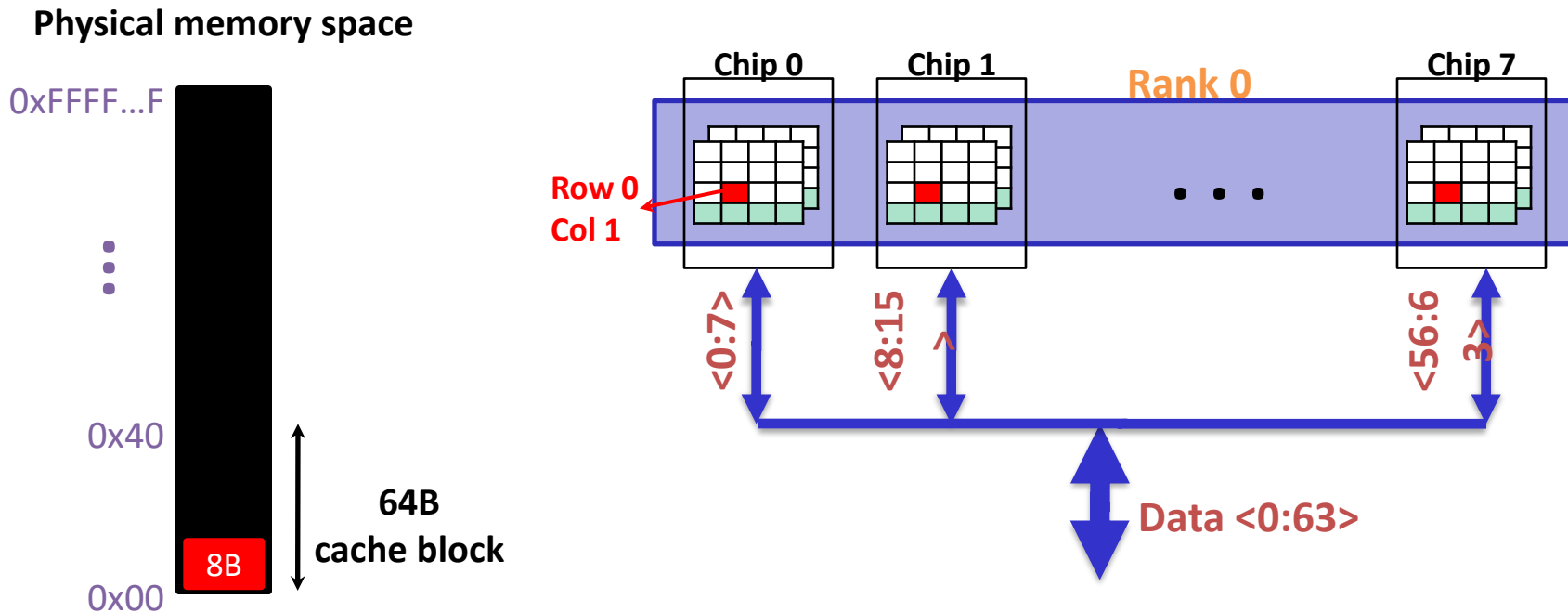
Transferring a cache block



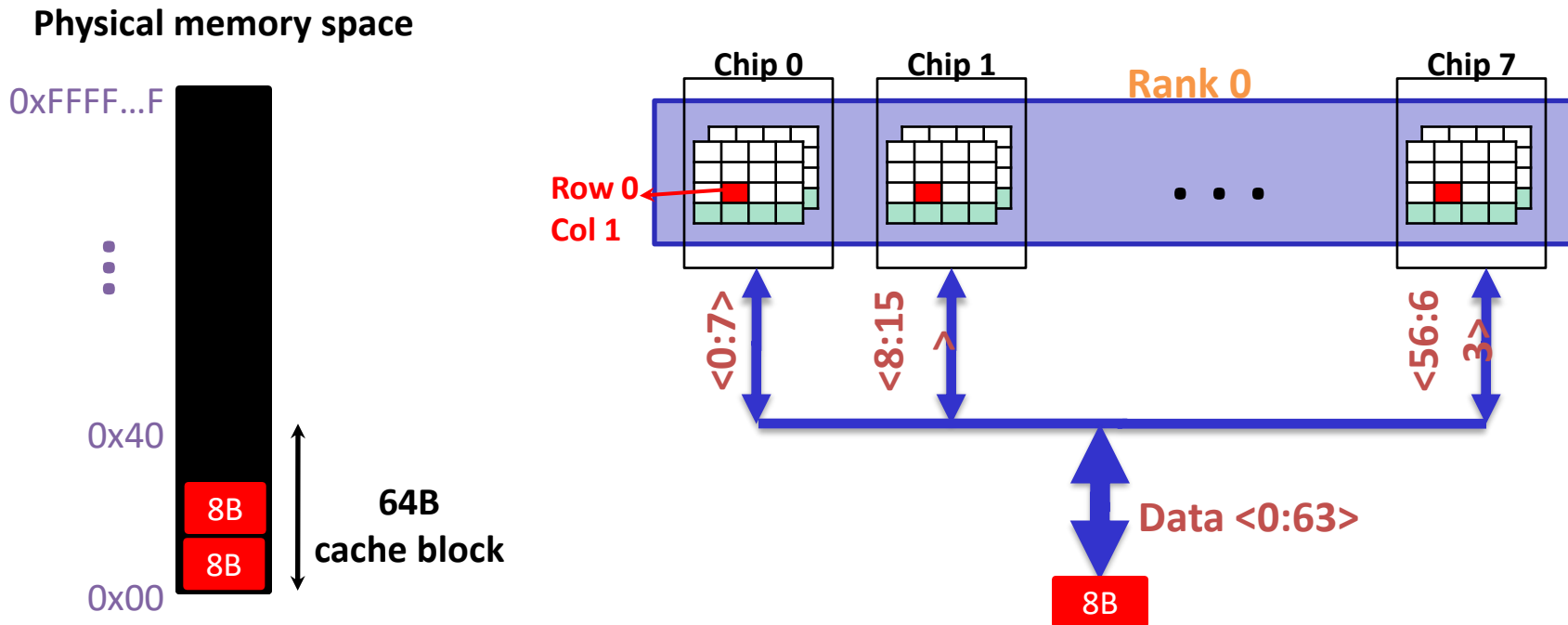
Transferring a cache block



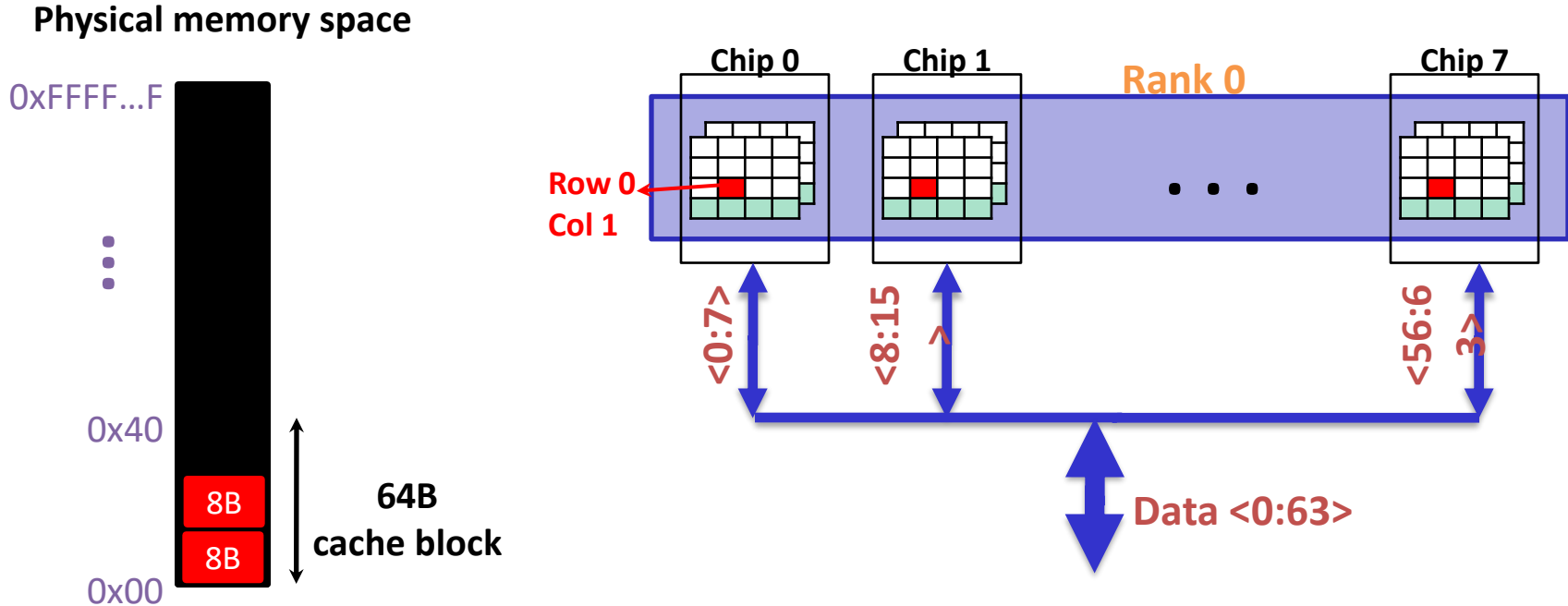
Transferring a cache block



Transferring a cache block



Transferring a cache block



A 64B cache block takes 8 I/O cycles to transfer.

During the process, 8 columns are read sequentially.



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